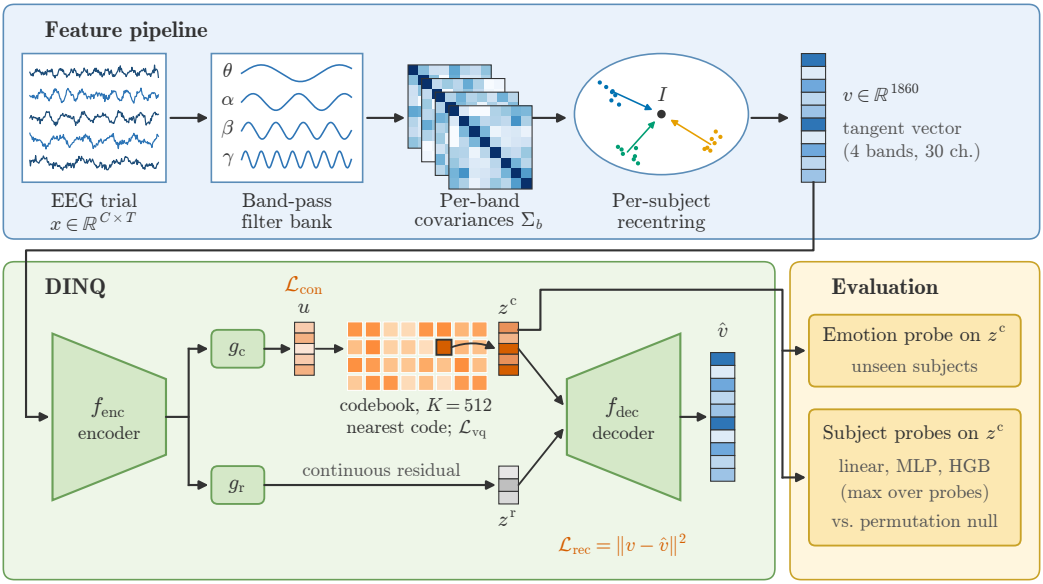
Feature pipeline
EEG trial
$x \in \mathbb{R}^{C \times T}$
Band-pass
filter bank
θ
α
β
γ
Per-band
covariances Σ_b
I
Per-subject
recentring
$v \in \mathbb{R}^{1860}$
tangent vector
(4 bands, 30 ch.)
DINQ
f_{enc}
encoder
g_{c}
g_{r}
$\mathcal{L}_{\text{con}}$
u
codebook, $K = 512$
nearest code; $\mathcal{L}_{\text{vq}}$
z^{c}
continuous residual
z^{r}
f_{dec}
decoder
$\hat{v}$
$\mathcal{L}_{\text{rec}} = \|v - \hat{v}\|^2$
Evaluation
Emotion probe on z^{c}
unseen subjects
Subject probes on z^{c}
linear, MLP, HGB
(max over probes)
vs. permutation null